Frederik Spors
DS Capstone - Bibliography
09/13/2026

Annotated Bibliography

Source of Data: Dataset from Professor Harris and she didn't share it with me yet.

Source 1.

Dmochowski, J. P., Sajda, P., Dias, J., & Parra, L. C. (2012). Correlated components of ongoing EEG point to emotionally laden attention – a possible marker of engagement? Frontiers in Human Neuroscience, 6, 112.

Github tied to this paper: https://github.com/ML-D00M/ISC-Inter-Subject-Correlations/tree/main

- This paper uses Correlated Components Analysis which is the method behind most EEG work for inter-subject correlation (ISC). Instead of comparing electrodes one at a time, it finds a weighted combination of all the channels that is as similar as possible across two recordings and these can come from one person watching a clip twice or from two people watching it once. Participants viewed short film clips and their EEG data was recorded, and correlation was measured and tested against a shuffled baseline and corrected for multiple comparisons. Correlation was highest during suspenseful moments, dropped on a second viewing, and decreased further when the scenes were played out of order. For my project, the analysis would follow similar logic, comparing coherent versus scrambled comparison. The dots were shuffled so that the low-level visual input stays identical and only the meaningful structure is removed, so a drop in correlation would point to higher-level processing.

Source 2.

Harris, A. M., Denaro, C. M., & Reed, C. L. (2025). Aperiodic activity masks sensorimotor mu suppression during human action observation. Social Cognitive and Affective Neuroscience, 20(1), nsaf059.

- This study tested whether the mu rhythm decreases when people watch others actions. Coherent movement showed movement that mimics that of a human, while the scrambled versions randomized the dots' starting positions and movement, not mimicking a human. Finger tapping produced clear mu suppression, but the coherent versus scrambled comparison did not, until the aperiodic component (the background activity) was removed. A higher aperiodic component was masking the effect, which even after correction was small (d = 0.25). My capstone project will use the same stimuli and the same coherent versus scrambled contrast. The difference is that they compare average mu, but I will see whether the time course of the response is shared across participants using inter-subject correlation.

Source 3.

Keysers, C., Gazzola, V., & Wagenmakers, E.-J. (2020). Using Bayes factor hypothesis testing in neuroscience to establish evidence of absence. Nature Neuroscience, 23(7), 788–799.

- This paper is also important because it talks about why standard p-values cannot tell the difference between a result showing that an effect is absent and a result that is simply uninformative. If a p-value is non-significant it only means the data failed to reject the null, not that the null is supported. Bayes factors can help with this because it compares how well the null and the alternative each predict the observed data, producing a number that can favor either one. A Bayes factor near 1 means the data are inconclusive, while values clearly above or below 1 give evidence for one hypothesis over the other. This is important for my project because it could be that there is no coherent versus scrambled mu effect with standard analysis and if my ISC analysis also shows no difference between the conditions, a p-value cannot tell if there is no effect or if the data is too noisy. Bayes factor can help me report on this distinction.